

Master 2 Internship

Deep Priors for the Deconvolution of Electric field with GRAND.

Simon Prunet ¹ Sara El bouch ²

¹ Dr, CNRS, J.-L Lagrange simon.prunet@oca.eu

²MCF, Université Côte d’Azur, J.-L Lagrange sara.el-bouch@univ-cotedazur.fr

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1 General Context

Giant Radio Array for Neutrino Detection (GRAND)¹ project aims to detect ultra-high energy cosmic rays, gamma rays, and neutrinos by observing the radio emission produced in extensive air showers in the atmosphere. Each of the 20 giant antenna arrays of 10,000 square kilometer each, will record a voltage trace $\mathbf{v}(t) \in \mathbb{R}^3$ that results from the true electric field $\mathbf{e}(t) \in \mathbb{R}^3$ of the radio pulse filtered by the effective length of the antenna $\mathbf{L}_{\text{eff}}(\theta, \phi, f)$, and their radio-frequency (RF) chain denoted $\mathbf{T}(f)$:

$$\mathbf{v}(t) = \mathcal{F}^{-1} \left(\underbrace{\mathbf{T}(f) \cdot \mathbf{L}_{\text{eff}}(\theta, \phi, f)}_{\mathbf{H}(f) \in \mathbb{R}^{3 \times 3}} \cdot \mathcal{F}(\mathbf{e}(t)) \right) + \mathbf{n}(t) \quad (1)$$

where $\mathcal{F}, \mathcal{F}^{-1}$ denote respectively the Fourier transform and its inverse, θ denotes the angle between the shower axis and the vertical, while the azimuth angle ϕ is measured from geographic North, and $\mathbf{n}(t)$ additive noise dominated by Galactic emissions at GRAND frequencies.

Recovering $\mathbf{e}(t)$ from the noise-corrupted voltages is an ill-posed problem as the frequencies outside the bandwidth of $\mathbf{H}(f)$ are strongly attenuated and further corrupted by noise. Classical approaches rely mainly on regularized least squares. Even-though these methods are optimal under the hypothesis of Gaussian sources, this assumption fails when working with real data. Moreover, the GRAND project has access to ZHaires simulations [1]. These simulations provide realistic radio pulses in atmospheric air shower at the antenna level.

¹<https://grand.cnrs.fr>

1.1 Proposed Methodology

For these reasons, we choose to leverage learned priors using neural network architectures. The latter are trained on a *denoising proxy task*, as a means to learn a (gradient) of the prior on the pulses provided by the training set of simulations. This paradigm has been shown to substantially outperform the hand-crafted regularizers when *plugged* into inverse-problem frameworks. The taxonomy of these methods divides them into two main categories: **Plug and Play** (PnP) priors such as [5] and **Regularization by Denoising** (RED) methods [4].

We have conducted a sanity check on a synthetic toy example to validate the use of PnP methods for this problem. We have trained a deep denoiser termed RefineNet, inspired by [3], a variant of a U-Net architecture schematized in 1, to remove Gaussian noise and validated that it learns a meaningful prior on the data. We have also tested the splitting algorithm in [5] to verify that we can deconvolve the traces when using an ideal bandpass $\mathbf{H}(f)$. Thus, the candidate will not start from scratch and will have access to this repository in https://github.com/Saraoid4/PnP_Deconvolution.

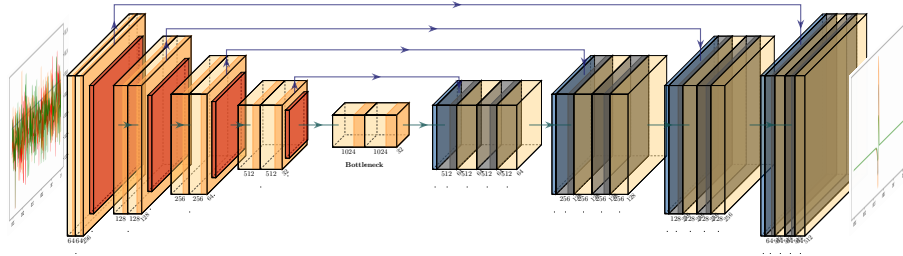


Figure 1: Deep denoiser trained on Gaussian-corrupted synthetic pulses.

2 Internship Objectives

The core of this internship will primarily focus on the ZHaires simulation data.

- Some traces in the simulation data are very noisy beyond 70 MHz, and this probably hinders the learning of a meaningful prior; thus, pre-processing the simulation data prior to learning is necessary.
- After this pre-processing step, the first objective is to train a suitable deep denoiser. Even though neural networks provide good denoising capability, they lack convergence guarantees when integrated into inverse-problems. A possible denoiser to tackle this shortcoming was proposed in [2].
- Once the training is complete. The deep denoiser will be plugged into a classical inverse-problem algorithm, with a proper choice of the set of

hyperparameters. The forward model $\mathbf{H}(f)$ in (1) is readily available in https://github.com/Saraoid4/PnP_Deconvolution.

- Finally, the method should be evaluated not only in terms of convergence but also in terms of in-band and out-of-band reconstruction of the electric fields, and benchmarked against regularized least squares methods to highlight the importance of this methodology.

3 Required skills

This internship targets Master 2 students with a specialty in Applied Mathematics, Signal processing/Data processing. The candidate should be at ease with:

- Statistical Signal Processing, Fourier analysis, inverse problems.
- Modern Neural Network architectures, as well as a Python framework for their implementation (preferably PyTorch/ PyTorch lightning).
- An interest in astroparticle physics.

4 Supervision and environment

The internship will take place in the J.-L Lagrange laboratory (UMR 7293), a joint laboratory between Université Côte d’Azur, CNRS and Observatoire de la Côte d’Azur based in Nice, France. She/He will be part of the Signal&Image team, and mainly supervised by:

Supervisors:

- Simon prunet, simon.prunet@oca.eu
- Sara El Bouch sara.el-bouch@univ-cotedazur.fr

Duration: 5 to 6 months internship, provisional starting date February 2027.

Location: Bâtiment Fizeau, 28 Av. Valrose Parc Valrose, 06108 Nice

References

- [1] Jaime Alvarez-Muniz, Washington R Carvalho Jr, and Enrique Zas. “Monte Carlo simulations of radio pulses in atmospheric showers using ZHAireS”. In: *Astroparticle Physics* 35.6 (2012), pp. 325–341.
- [2] Samuel Hurault et al. “A relaxed proximal gradient descent algorithm for convergent plug-and-play with proximal denoiser”. In: *International Conference on Scale Space and Variational Methods in Computer Vision*. Springer. 2023, pp. 379–392.

- [3] Guosheng Lin et al. “Refinenet: Multi-path refinement networks for high-resolution semantic segmentation”. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017, pp. 1925–1934.
- [4] Edward T Reehorst and Philip Schniter. “Regularization by denoising: Clarifications and new interpretations”. In: *IEEE transactions on computational imaging* 5.1 (2018), pp. 52–67.
- [5] Kai Zhang et al. “Plug-and-play image restoration with deep denoiser prior”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44.10 (2021), pp. 6360–6376.